



Girraween High School

Year 12 Half Yearly Examination 2017

Mathematics Extension 2

Time Allowed-Two hours (Plus 5 minutes reading time)

Instructions:

*Attempt all questions.

*Show all necessary working.

*Marks may be deducted for careless or badly arranged work.

*Board – approved calculators may be used

1. The polynomial $P(x)$ with real coefficients has $x = 1$ as a root of multiplicity 2 and $x + i$ as a factor. Which one of the following expressions could be a factorised form of $P(x)$?

A) $(x^2 + 1)(x - 1)^2$ B) $(x + i)^2(x - 1)^2$ C) $(x - i)^2(x - 1)^2$ D) $(x^2 + 1)(x - i)^2$

2. What are the foci of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$?

A) $((0, \pm 3))$ B) $(\pm 5, 0)$ C) $(\pm 3, 0)$ D) $(0, \pm 5)$

3. Given that $z = 1 + i$, what is the value of z^8 ?

A) -16 B) -8 C) 8 D) 16

4. The equation $x^4 + px + q = 0$, where $p \neq 0$ and $q \neq 0$, has roots α, β, γ and δ . What is the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$?

A) $-4q$ B) $p^2 - 2q$ C) $p^4 - 2q$ D) p^4

5. The sum of the eccentricities of two different conics is $\frac{3}{4}$.

Which pair of conics could this be?

A) Circle and ellipse B) Ellipse and parabola C) Parabola and hyperbola D) Hyperbola and circle

Question 6. (21 marks)

a) If $z = 1 + 2i$, find in the form $a + ib$, where a and b are real, the values of

i) $iz + \bar{z}$ 1

ii) $\frac{1}{z}$ 1

b) The points A and C represent the complex numbers $1 + i$ and $7 + 3i$ respectively.
Find the complex number ω , represented by B such that $\triangle ABC$ is isosceles
and right angled at B . 3

c) Shade on an Argand diagram the region represented by the complex number z
where $\frac{\pi}{4} \leq \arg z \leq \pi$, $1 \leq \operatorname{Im}(z) \leq 3$ and $|z| \leq 4$ 3

d) Let $\alpha = 1 + i\sqrt{3}$ and $\beta = 1 + i$

i) Find $\frac{\alpha}{\beta}$, in the form $x + iy$. 1

ii) Express α in modulus-argument form. 1

iii) Given that β has the modulus-argument form $\beta = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$,
find the modulus-argument form of $\frac{\alpha}{\beta}$. 2

iv) Hence find the exact value of $\sin \frac{\pi}{12}$. 1

e) i) Using De Moivre's theorem, find an expression for $\cos 3\theta$ in terms of $\cos \theta$. 3

ii) If $\cos 3\theta = -\frac{1}{\sqrt{2}}$, show that $x = 2\sqrt{2} \cos \theta$ satisfies the equation $x^3 - 6x + 4 = 0$. 2

iii) Hence, find the exact values of the roots of this equation. 3

Question 7. (18 marks)

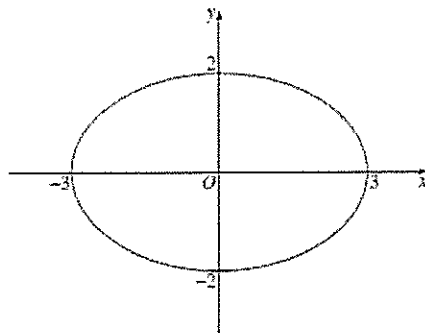
- a) The cubic polynomial $x^3 + 2x^2 - 5x - 6 = 0$ has roots α, β and γ . Find the values of
- i) $\alpha + \beta + \gamma$ 1
 - ii) $\alpha^2 + \beta^2 + \gamma^2$ 2
 - iii) $\alpha^3 + \beta^3 + \gamma^3$ 2
- b) If $2 + i$ is a root of $P(x) = x^4 - 6x^3 + 9x^2 + 6x - 20$, resolve $P(x)$ into linear factors over the field of complex numbers. 4
- c) The equation $x^3 + 2x + 1 = 0$ has roots α, β and γ . Find in expanded form the monic cubic equation with roots $\alpha + 1, \beta + 1$ and $\gamma + 1$. 2
- d) i) Prove that the condition for the roots of $x^3 - px^2 + qx - r = 0$ to be in arithmetic progression is $2p^3 - 9pq + 27r = 0$. 4
- ii) Hence solve the equation $x^3 - 12x^2 + 39x - 28 = 0$ 3
- e) i) If $x^3 + px + q$ has a double zero, prove that the double zero must be $-\frac{3q}{2p}$, $p \neq 0$ 2
- ii) Show also that if $x^3 + px + q$ has a double zero, then $4p^3 + 27q^2 = 0$ 2

Question 8 (15 marks)

- a) Find the equation of the tangent to the curve $x^2 - xy + y^3 = 1$ at the point $P(1,1)$ on the curve.

3

- b) The diagram below shows an ellipse.



- i) Write an equation for the ellipse. 1
- ii) Find the eccentricity of the ellipse. 1
- iii) Write the coordinates of the foci of the ellipse. 1
- iv) Write the equations of the directrices of the ellipse. 1

- c) i) Show that the equations of the tangent and the normal at $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

are $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ and $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$ respectively. 4

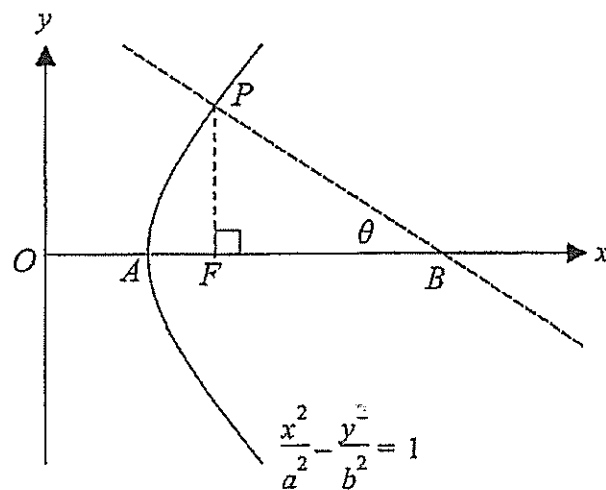
- ii) The tangent at P meets the x -axis at T . PN is perpendicular to the x -axis, and the normal at P meets the x -axis at G . Show that $OT \times NG = b^2$, where O is the centre of the ellipse. 4

Question 9 (10 marks)

- a) Sketch the hyperbola with equation $\frac{x^2}{64} - \frac{y^2}{36} = 1$, showing the foci, the directrices and the asymptotes.

4

b)



In the diagram, F is a focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with eccentricity e .

This branch of the hyperbola cuts the x -axis at A where $AF = h$. P is the point on the

hyperbola vertically above F and the normal at P cuts the x -axis at B making an

acute angle θ with the x -axis. (Assume that the equation of the normal at $(P(x_1, y_1))$ is given by

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2. \text{ Do not prove this)}$$

- i) Show that $\tan \theta = \frac{1}{e}$ 3

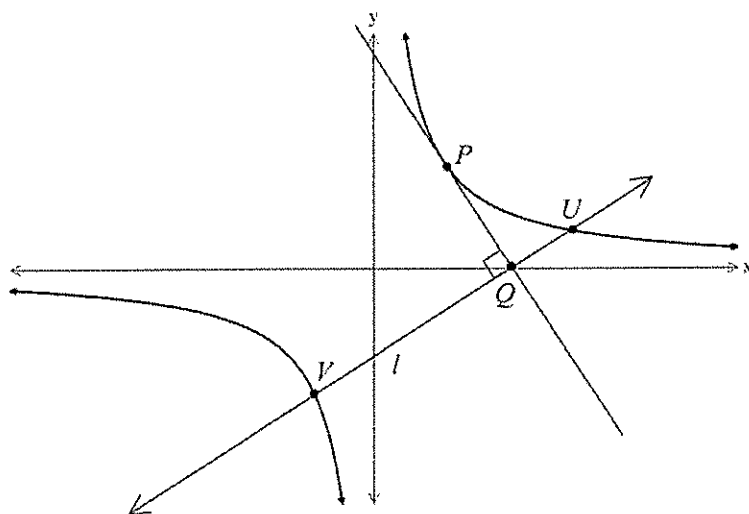
- ii) Show that $PF = h(e+1)$ 3

Question 10. (9 marks)

A tangent is drawn at any point $P\left(ct, \frac{c}{t}\right)$ on the hyperbola $xy = c^2$. This tangent

meets the x axis at Q . Through Q a straight line l is drawn perpendicular to the tangent.

The line l cuts the hyperbola in the two points U and V .



i) Show that the equation of the tangent is $x + t^2y = 2ct$ 2

ii) Find the coordinates of Q 1

iii) Find the equation of line l . 1

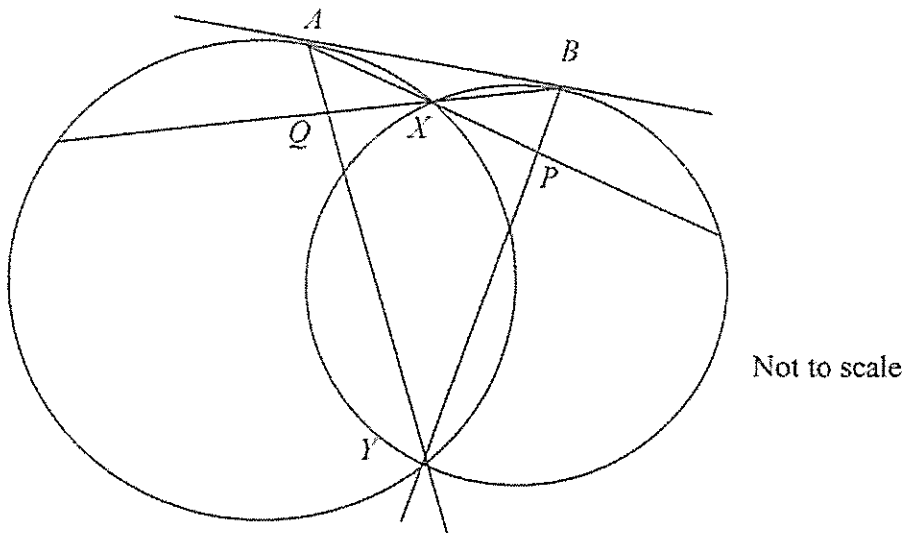
iv) If M is the midpoint of the interval UV , show that the coordinates of M are $(ct, -ct^3)$. 3

v) Hence find the locus of M as the point P varies. 2

Question 11 (7 marks).

Two circles intersect at X and Y and AB is a common tangent.

The lines AX and BY meet at P . The lines AY and BX meet at Q as shown in the diagram below.



- i) Copy the diagram in your answer booklet and show that $XPYQ$ is a cyclic quadrilateral, giving reasons.

4

- ii) Prove that $QP \parallel AB$.

3

Task 2 SOLUTIONS.

Multiple choice Questions.

① $(x+i)$ is a factor. $\therefore (x-i)$ is also a factor

$$(x+i)(x-i) = x^2 + 1$$

and root of multiplicity 2 $\Rightarrow (x-i)^2$

$\therefore$ (A)

② $\frac{x^2}{9} - \frac{y^2}{16} = 1 \Rightarrow a=3 \text{ and } b=4.$

$$b^2 = a^2(e^2 - 1)$$

$$16 = 9(e^2 - 1)$$

$$\therefore e = \frac{5}{3}$$

③ $\therefore \text{Foci} = (\pm 5, 0)$

④ $1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$

$$(1+i)^8 = 16 (\cos 2\pi + i \sin 2\pi) = 16$$

(D)

⑤ $x^4 + px + q = 0$ and $\alpha, \beta, \gamma, \delta$ are roots.

$$\therefore \alpha^4 = -p\alpha - q \quad \therefore \alpha + \beta + \gamma + \delta = 0$$

$$\beta^4 = -p\beta - q$$

$$\gamma^4 = -p\gamma - q$$

$$\delta^4 = -p\delta - q$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = -p(\alpha + \beta + \gamma + \delta) - 4q$$

$$= -4q$$

(A)

⑥ Circle has $e=0$, ellipse has $e<1$.
parabola has $e=1$, hyperbola has $e>1$

$\therefore$ (A)

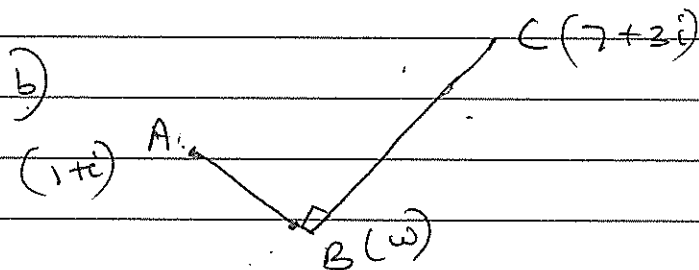
Question 6

a) (i) $z = 1+2i$

$$\therefore iz + \bar{z} = i(1+2i) + 1-2i$$

$$= i + 2 + 1 - 2i = -1 - i \quad (1)$$

(ii) $\frac{1}{\bar{z}} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{1-2i}{1^2+2^2} = \frac{1}{5} + \left(\frac{-2}{5}\right)i$ (1)



$$\vec{BA} = \vec{BC} \cdot i$$

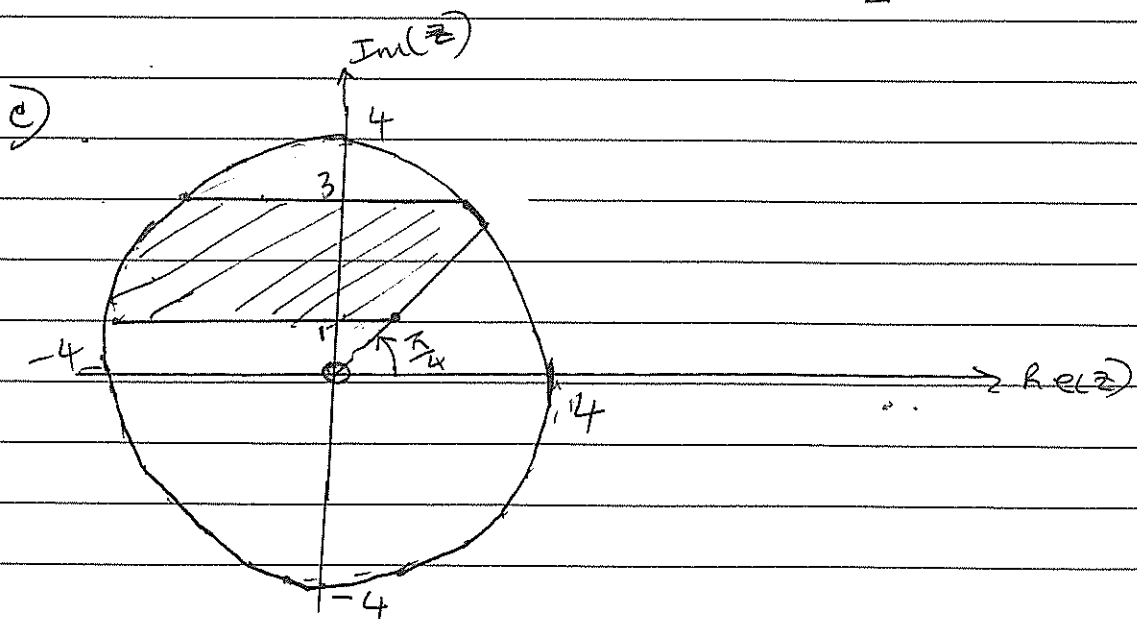
$$1+i-w = (7+3i-w)i$$

$$1+i-w = 7i-3-wi$$

$$w(1-i) = 1+i-7i+3 = 4-6i$$

$$\therefore w = \frac{4-6i}{1-i} = \frac{4-6i}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{4+6-2i}{2} = \underline{\underline{5-i}} \quad (2)$$



d) $\alpha = 1 + \sqrt{3}i$
 $\beta = 1 + i$

(i) $\therefore \frac{\alpha}{\beta} = \frac{1 + \sqrt{3}i}{1 + i} = \frac{1 + \sqrt{3}i}{1 + i} \times \frac{1 - i}{1 - i}$

$$= \frac{1 + \sqrt{3} + i(\sqrt{3} - 1)}{2}$$

(1)

(ii) $\alpha = 1 + \sqrt{3}i = 2 \cos \frac{\pi}{3}$

(1)

Given

(iii) $\beta = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$,

$$\frac{\alpha}{\beta} = \frac{2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)}{\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)} = \sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

(2)

(iv) $\frac{\alpha}{\beta} = \frac{1 + \sqrt{3} + i(\sqrt{3} - 1)}{2} = \sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$

equating the imaginary parts;

$$\frac{\sqrt{3} - 1}{2} = + \sqrt{2} \sin \frac{\pi}{12}$$

$$\therefore \sin \frac{\pi}{12} = + \left(\frac{\sqrt{3} - 1}{2\sqrt{2}} \right) = \frac{-1 + \sqrt{3}}{2\sqrt{2}}$$

(1)

d) (i) $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$, using

Let $c = \cos \theta$, $s = \sin \theta$

De Moivre's Theorem

$$\text{LHS} \Rightarrow c^3 + 3c^2(is) + 3c(is)^2 + (is)^3$$

$$= (c^3 - 3cs^2) + i(3c^2s - s^3)$$

equating real parts $\Rightarrow c^3 - 3cs^2 = \cos 3\theta$
 $= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$

$$\therefore \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

Question 6 continued.

$$(ii) \cos 3\theta = 4\cos^3\theta - 3\cos\theta = -\frac{1}{\sqrt{2}}$$

substitute $\cos\theta = \frac{x}{2\sqrt{2}}$ gives,

$$\frac{4x^3}{16\sqrt{2}} - \frac{3x}{2\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

$$\therefore 4x^3 - 24x = -16 \Rightarrow x^3 - 6x + 4 = 0.$$

$\therefore x = 2\sqrt{2}\cos\theta$ satisfies the equation
 $x^3 - 6x + 4 = 0.$

$$(iii) \cos 3\theta = -\frac{1}{\sqrt{2}} \Rightarrow 3\theta = \pm \frac{3\pi}{4} + 2k\pi, k = 0, 1$$

$$\therefore \theta = \frac{(8k \pm 3)\pi}{12}, k = 0, 1$$

$$= \frac{\pi}{4}, \frac{5\pi}{12}, \frac{11\pi}{12}.$$

Question 7

$x^3 + 2x^2 - 5x - 6 = 0$ has roots α, β and γ .

a) (i) $\alpha + \beta + \gamma = \boxed{-2}$

①

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= (-2)^2 - 2(-5) = 14$

②

(iii) $\alpha^3 + \beta^3 + \gamma^3 = ?$

α is a root $\Rightarrow \alpha^3 = -2\alpha^2 + 5\alpha + 6$

β " " $\Rightarrow \beta^3 = -2\beta^2 + 5\beta + 6$

γ " " $\Rightarrow \gamma^3 = -2\gamma^2 + 5\gamma + 6$

$$\begin{aligned}\alpha^3 + \beta^3 + \gamma^3 &= -(\alpha^2 + \beta^2 + \gamma^2) + 5(\alpha + \beta + \gamma) + 18 \\ &= (-2 \times 14) + 5(-2) + 18 \\ &= -28 - 10 + 18 = -20\end{aligned}$$

③

b) $P(x) = x^4 - 6x^3 + 9x^2 + 6x - 20$

If $(2+i)$ is a root, then $(2-i)$ is also a root
($\because P(x)$ has real coefficients)

Let $z = 2+i$ & $\bar{z} = 2-i$

$z + \bar{z} = 4$ and $z\bar{z} = 5$

$\therefore$ The quadratic equation with roots $(2+i), (2-i)$ is $x^2 - 4x + 5$

Divide $P(x)$ by $x^2 - 4x + 5$ to find other roots

$$\begin{array}{r} x^2 - 4x + 5 \overline{) x^4 - 6x^3 + 9x^2 + 6x - 20} \\ \underline{x^4 - 4x^3 + 5x^2} \\ -2x^3 + 4x^2 + 6x \\ \underline{-2x^3 + 8x^2 + 10x} \\ -4x^2 + 16x - 20 \\ \underline{-4x^2 + 16x - 20} \\ 0 \end{array}$$

$\therefore P(x) = (x^2 - 4x + 5)(x^2 - 2x - 4)$

$= [x - (2+i)][x - (2-i)]$

$(x - 1 - i\sqrt{5})(x - 1 + i\sqrt{5})$

④

c) $x^3 + 2x + 1 = 0$ has roots α, β and γ .

Let $y = x + 1 \therefore x = y - 1$.

sub. in $x = y - 1$ into $x^3 + 2x + 1 = 0$ gives.

$$(y-1)^3 + 2(y-1) + 1 = 0$$

$$\therefore y^3 - 3y^2 + 5y - 2 = 0$$

$$\therefore x^3 - 3x^2 + 5x - 2 = 0$$

(2)

d) (i) $x^3 - px^2 + qx - r = 0$ — (1)

let the roots be $a-d, a, a+d$

$$\therefore \text{sum of roots} \Rightarrow 3a = p$$

$$\therefore a = \frac{p}{3}$$

sub. in $\frac{p}{3}$ into (1) as a is a root of (1)

$$\Rightarrow \frac{p^3}{27} - p \cdot \frac{p^2}{9} + q \cdot \frac{p}{3} - r = 0$$

$$p^3 - 3p^3 + 9pq - 27r = 0$$

$$\therefore 2p^3 - 9pq + 27r = 0$$

(ii) solve $x^3 - 12x^2 + 39x - 28 = 0$ — (2)

equating coefficients $\Rightarrow p = 12 \therefore a = \frac{12}{3} = 4$

(1) and (2)

$$\therefore (x-4) \text{ is a root of (2)}$$

(3/2)

Question 7 continued

$$\begin{array}{r} x^2 - 8x + 7 \\ (ii) \quad x-4 \overline{) x^3 - 12x^2 + 39x - 28} \\ \underline{x^3 - 4x^2} \\ -8x^2 + 39x \\ \underline{-8x^2 + 32x} \\ 7x - 28 \end{array}$$

$$\therefore x^3 - 12x^2 + 39x - 28 = (x-4)(x-7)(x-1)$$

e) Let $f(x) = x^3 + px + q$. — (1)

$$f'(x) = 3x^2 + p$$

$$\text{double root} \Rightarrow f'(x) = 0 \Rightarrow x^2 = -\frac{p}{3}$$

$$\text{sub. in } x^2 = -\frac{p}{3} \text{ into } f(x) = 0$$

$$\therefore x \cdot \left(-\frac{p}{3}\right) + px + q = 0$$

$$\therefore -\frac{px}{3} = -q$$

$$\therefore x = -\frac{3q}{2p}$$

This value of $x = -\frac{3q}{2p}$ satisfies both

$$f(x) = 0 \text{ and } f'(x) = 0 \quad (\text{double root})$$

$$\therefore \text{sub. in } x = -\frac{3q}{2p} \text{ into } f'(x) = 0$$

$$3 \cdot \left(-\frac{3q}{2p}\right)^2 + p = 0 \Rightarrow \frac{27q^2}{4p^2} + p = 0$$

$$\therefore 27q^2 + 4p^3 = 0$$

Question 8

a) $x^2 - xy + y^3 = 1.$

Differentiating with respect to x gives,
 $2x - xy' - y + 3y^2 \cdot y' = 0$

$$2x - y + y'(3y^2 - x) = 0$$

$$\therefore y' = \frac{y - 2x}{3y^2 - x}$$

$$\text{at } (1, 1), y' = \frac{1 - 2}{3 - 1} = -\frac{1}{2}$$

$\therefore$ equation! $y - 1 = -\frac{1}{2}(x - 1)$

$$2y - 2 = -x + 1 \Rightarrow x + 2y - 3 = 0$$

(3)

b) (i) $\frac{x^2}{9} + \frac{y^2}{4} = 1$ ($a=3, b=2$)

(ii) $b^2 = a^2(1 - e^2)$

$$\therefore 4 = 9(1 - e^2) \Rightarrow e^2 = 1 - \frac{4}{9} \Rightarrow e = \frac{\sqrt{5}}{3}$$

(iii) Foci = $(\pm ae, 0) = (\pm \sqrt{5}, 0)$

(iv) $\therefore x = \pm \frac{a}{e} \Rightarrow x = \pm \frac{9}{\sqrt{5}}$

Question 8 (continued)

c) (i) Differentiating both sides of

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{2x}{a^2} + \frac{2y}{b^2} y' = 0$$

$$\therefore y' = -\frac{b^2 x}{a^2 y}$$

at $P(x_1, y_1)$, the gradient is $= -\frac{b^2 x_1}{a^2 y_1}$,

$\therefore$ The equation of tangent is.

$$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$\therefore a^2 y y_1 - a^2 y_1^2 = -b^2 x x_1 + b^2 x_1^2$$

$$\therefore b^2 x x_1 + a^2 y y_1 = b^2 x_1^2 + a^2 y_1^2$$

$$\left(\frac{1}{a^2 b^2}\right) \quad \frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

(as (x_1, y_1) lie on the ellipse)

$$\therefore \frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1$$

(2)

(ii) The equation of the normal.

$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

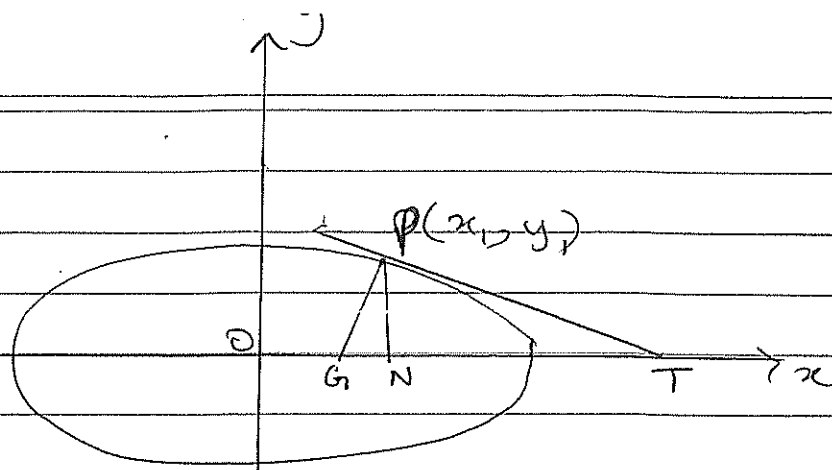
$$\therefore b^2 x_1 y - b^2 x_1 y_1 = a^2 x y_1 - a^2 x_1 y$$

$$a^2 y_1 x - b^2 x_1 y = x_1 y_1 (a^2 - b^2)$$

$$\left(\frac{1}{a^2 - b^2} x_1 y_1\right) \quad \frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(2)

(ii)



coordinates of T $\Rightarrow$ sub. in $y=0$

$$\therefore \frac{xx_1}{a^2} + 0 = 1 \Rightarrow x = \frac{a^2}{x_1}$$

$$\therefore OT = \frac{a^2}{x_1}$$

coordinates of G. $\Rightarrow$ sub. in $y=0$

$$\therefore \frac{a^2x}{x_1} = a^2 - b^2 \Rightarrow x = \frac{x_1(a^2 - b^2)}{a^2}$$

$$\therefore NG = x_1 - OG$$

$$= x_1 - \frac{x_1(a^2 - b^2)}{a^2} = \frac{x_1 b^2}{a^2}$$

$$\therefore OT \times NG = \frac{a^2}{x_1} \times \frac{x_1 b^2}{a^2} = b^2$$

$$b^2 = a^2(e^2 - 1)$$

$$\Rightarrow y_1^2 = \frac{b^2}{a^2} \cdot b^2$$

$$\therefore y_1 = \frac{b^2}{a}$$

$\therefore$ the gradient of normal at P is

$$= -a \cdot \frac{\frac{b^2}{a}}{b^2 e} = \frac{-b^2}{b^2 e} = \frac{-1}{e}$$

$$\therefore \tan \theta = \frac{1}{e}$$

(ii) The equation of normal at P:

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$\therefore$ coordinates of B $\Rightarrow$ ^{sub in} $y = 0$

$$\frac{a^2 \cdot x}{ae} = \frac{a^2 + b^2}{a} \Rightarrow x = \frac{(a^2 + b^2)e}{a}$$

$$\begin{aligned} \therefore FB &= OB - OF = \left(\frac{a^2 + b^2}{a} \right) e - ae \\ &= \frac{a^2 e + b^2 e - a^2 e}{a} = \frac{b^2 e}{a} \end{aligned}$$

$$\tan \theta = \frac{PF}{FB} = \frac{PF}{\frac{b^2 e}{a}} = \frac{1}{e} \quad (\text{from (i)})$$

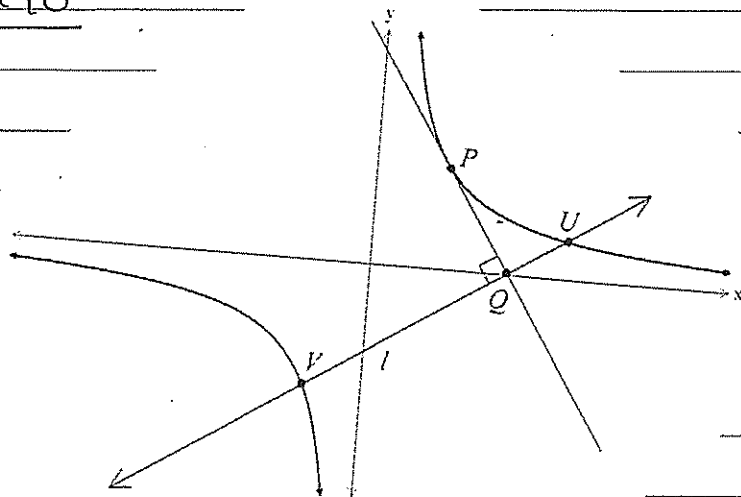
$$\therefore PF = \frac{\frac{b^2 e}{a} \times 1}{e} = \frac{b^2}{a}$$

$$AF = h \Rightarrow ae - a = h = a(e - 1)$$

$$\begin{aligned} \therefore PF &= \frac{b^2}{a} = \frac{a^2(e^2 - 1)}{a} = \frac{a^2(e-1)(e+1)}{a} \\ &= a(e-1)(e+1) = h(e+1) \end{aligned}$$

(3)

Question 10



$$xy = c^2$$

(i) Differentiating both sides give,

$$xy' + y = 0 \quad \therefore y' = -\frac{y}{x}$$

$$\therefore \text{gradient at } P = -\frac{c}{\frac{c}{t}} = -\frac{1}{t^2}$$

$\therefore$ equation of the tangent:

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$\therefore t^2 y - ct = -x + ct$$

$$\therefore x + t^2 y = 2ct \quad (2)$$

(ii) coordinates of Q $\Rightarrow$ sub-in $y=0$

$$\therefore x = 2ct$$

$$\therefore Q = (2ct, 0) \quad (1)$$

(iii) gradient of the normal at P $= t^2$

$$\therefore \text{equation of } l: y - 0 = t^2(x - 2ct)$$

$$y = t^2 x - 2ct^3 \quad (1)$$

To find the coordinates of v ,

(iv) $y = t^2x - 2ct^3 \quad \text{--- (1)}$

$xy = c^2 \quad \text{--- (2)}$

Solve simultaneously. (1) and (2)

sub. in (2) in (1)

$$\frac{c^2}{x} = t^2x - 2ct^3$$

$$\therefore c^2 = t^2x^2 - 2ct^3x$$

$$t^2x^2 - 2ct^3x - c^2 = 0.$$

$$\text{mid point of } UV = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{sum of roots} = x_1 + x_2 = -\frac{b}{a}$$

$$= \frac{2ct^3}{t^2} = 2ct$$

$$\therefore \text{mid point } m = \left(\frac{2ct}{2}, \frac{y_1 + y_2}{2} \right) = (ct, y_1)$$

To find y_1 , sub. in $x = ct$ in (1)

$$y = t^2 \cdot ct - 2ct^3 = -ct^3$$

$$\therefore m = (ct, -ct^3)$$

(3)

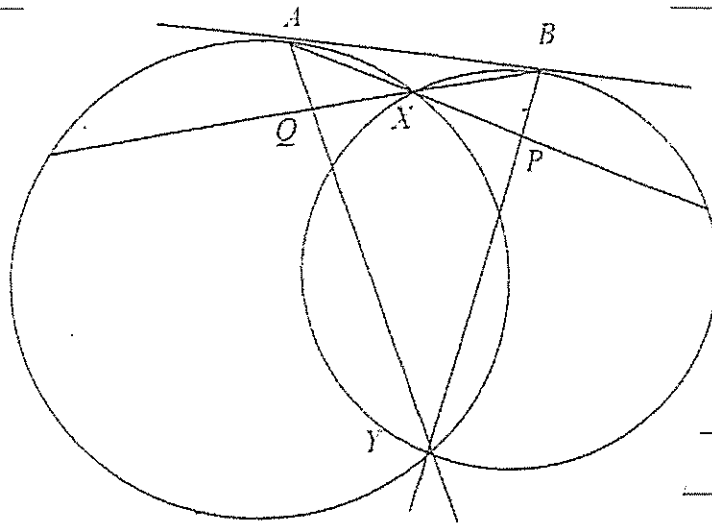
v) $x = ct \quad \therefore t = \frac{x}{c}$

$$y = -ct^3$$

$$= -c \cdot \frac{x^3}{c^3} = -\frac{x^3}{c^2}$$

(2)

Question 10



(i) Join XY . Let $\angle BAX = x$, $\angle ABX = y$

$\angle BXP = x + y$ (In $\triangle BAX$, exterior angle.
= sum of interior opposite $\angle$ s)

$\angle BAX = \angle AYX$ (angle between tangent
= x and chord = $\angle$ in alternate segment)

similarly,
 $\angle ABX = \angle BYX = y$

$\therefore \angle QYP = x + y = \angle BXP$

$\therefore XPYQ$ is a cyclic quadrilateral
(exterior $\angle$ = interior opposite $\angle$).

(4)

(ii) Join PQ .

$\angle XPQ = \angle QYX = x$ ($\angle$ at circumference
standing on the same
arc as $XPYQ$ is cyclic)

$\therefore \angle BAP = \angle XPQ = x$.

$\therefore AB \parallel PQ$ (alternate angles equal)

(3)